CANALRUNG · FEASIBILITY ASSESSMENT

CanalRung feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/canalrung		
Source	docs/08-feasibility.md @ ce4cb4e		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-05	Amish Chadha	First feasibility assessment (kit 1.9.1 upgrade) for option A, with a line on options B and C

CanalRung feasibility assessment

Option A is feasible on paper, with the rope margin, the cost (USD 605.20 against the USD 500 target) and the climb on a slick wall as the open points; none of it is shown by test. Method: STANDARDS section 23.

Table 1. Feasibility by aspect (option A unless stated).

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	The reach, mass, buoyancy, rung, rope and anchor numbers all meet their targets on paper; 13 STEM claims for A: 4 within, 9 plausible	High	Rope factor 4.08 against 4 (3.3 after ageing), the saturated-soil footing factor 1.59 against 1.5 (before a slope-crest reduction the CAL does not include) and the eye nut load off its axis are thin or unchecked	CNR-STM-001, CNR-CAL-001
Materials	Stock aluminium tube, galvanised SHS, EN 1891 rope, nylon, HDPE, closed-cell foam, concrete	Medium	UV ageing of rope and foam is a test item; aluminium on galvanised steel is a weak galvanic couple (S10, S11)	CNR-STM-001
Manufacturing and assembly	Cut, drill and bolt with hand tools; no welding; 178 parts, 135 made or bought and 43 fasteners in the D3 model	High	Whether a village workshop can drill and ream 26 rung ends to size	CNR-BLD-001, `cad/fidelity.json`
Power and controls	None	High	No electrical or electronic parts	none
Supply of parts	Catalogue items in most regions; the rope, eye nut and foam must declare their ratings	Medium	Prices are indicative and unverified on 2026-11-05 (see CNR-MKT-001)	BOM
Cost	USD 605.20 for the parts, USD 105.20 over the value-engineering target	Medium	The footing is USD 143 of it; no labour counted	BOM, CNR-CAL-001 I1
Use and safety	A bystander can throw it, a person can climb it	Low	Neither is shown; both need supervised trials (TRL 4) and a person must never enter the water	CNR-PRS-001 PR-002, PR-005, PR-007
Options B and C	B (10.6 kg rigid ladder) and C (hand winch with a floating sling) are buildable on paper at D3	Medium	B is hard to handle over an edge; C needs welding and an 8 mm rope rating that is not cited	CNR-OPT-001, CNR-STM-001

Ratings: High, Medium, Low.